

Multiple Nmap scanning techniques were performed against the authorized scanme.nmap.org testing host, including TCP port scanning, UDP port scanning, service/version detection, and Nmap Scripting Engine (NSE) scanning. The results were analyzed to identify accessible services and gather information about the target system.

The Nmap assessment was limited to scanme.nmap.org, an Nmap-provided target intended for security scanning demonstrations. The assessment was limited to the scanning and enumeration activities documented in this report.

Item	Configuration
Target	scanme.nmap.org
IPv4 address observed	45.33.32.156
Tool	Nmap 7.94
Scanner	Kali Linux
Testing activities	TCP scanning, UDP scanning, service enumeration, NSE scripts

Nmap scan configuration

Scan	Command	Purpose
TCP	nmap scanme.nmap.org	Identify commonly accessible TCP ports
UDP	nmap -sU scanme.nmap.org	Identify UDP services
Service detection	nmap -sV scanme.nmap.org	Identify service/version information
NSE scripts	nmap -sC scanme.nmap.org	Run Nmap's default scripts for additional enumeration

TCP Port Scanning

```

(root@kali)-[/home/kali]
# nmap scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:33 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.086s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered   smtp
80/tcp    open       http
139/tcp    filtered   netbios-ssn
445/tcp    filtered   microsoft-ds
9929/tcp   open       nping-echo
31337/tcp  open       Elite

Nmap done: 1 IP address (1 host up) scanned in 4.15 seconds

```

The initial TCP scan identified multiple ports in different states. Ports 22/tcp and 80/tcp were identified as open and associated with SSH and HTTP respectively. Ports 9929/tcp and 31337/tcp were also identified as open. Ports 25/tcp, 139/tcp, and 445/tcp were reported as filtered, meaning Nmap could not establish that those ports were openly accessible.

UDP Port Scanning

```

(root@kali)-[/home/kali]
# nmap -sU scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:34 PDT
Warning: 45.33.32.156 giving up on port because retransmission cap hit (10).
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.082s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 closed udp ports (port-unreach)
PORT      STATE      SERVICE
68/udp    open|filtered  dhcp
123/udp    open        ntp
135/udp    filtered     msrpc
136/udp    filtered     profile
137/udp    filtered     netbios-ns
138/udp    filtered     netbios-dgm
139/udp    filtered     netbios-ssn
445/udp    filtered     microsoft-ds
1900/udp   open|filtered  upnp

Nmap done: 1 IP address (1 host up) scanned in 1155.64 seconds

```

A UDP scan was performed using the “-sU” option. The scan identified UDP/123 as open and associated it with NTP. Several other ports were reported as filtered or open|filtered. UDP

scanning took significantly longer than the TCP scan, completing in approximately 1155 seconds. The UDP scan generated a retransmission warning, indicating that Nmap reached its retransmission limit for at least one port. This limitation was considered when interpreting the UDP results, particularly ports reported as “open|filtered”.

Service and Version Enumeration

```
(root@kali)~[/home/kali]
# nmap -sV scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:36 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.087s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.22 seconds
```

Service/version detection was performed using Nmap's “-sV” option. The scan identified OpenSSH on port 22 and Apache HTTP Server on port 80. Nmap also identified an Nping echo service on port 9929. Port 31337 was identified as “tcpwrapped”, indicating that Nmap could not determine a more specific service through this scan. The identified software versions provide information that can be used during subsequent vulnerability assessment and verification.

NSE Script Enumeration

```

(root@kali)-[/home/kali]
# nmap -sC scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-21 08:42 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.10s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open  http
|_ http-favicon: Nmap Project
|_ http-title: Go ahead and ScanMe!
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 7.33 seconds

```

The Nmap Scripting Engine was used with the “-sC” option to perform additional enumeration. The scripts returned SSH host-key information and HTTP metadata, including the HTTP title and favicon information. This additional information supplements the basic port and service enumeration performed by the previous scans.

Nessus Vulnerability scan

The screenshot displays the Tenable Nessus Essentials web interface. The main panel shows a table of scan results for the host 'scanme.nmap.org'. The table has columns for Host, Auth, and Vulnerabilities. The scan status is 'Fail' with a score of 33. The right-hand panel provides 'Scan Details' including the policy, status, severity base, scanner, start/end times, and elapsed time. Below the details is a 'Vulnerabilities' section with a donut chart showing the distribution of vulnerabilities by severity: Critical (red), High (orange), Medium (yellow), Low (green), and Info (blue).

Host	Auth	Vulnerabilities
scanme.nmap.org	Fail	33

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 9:06 AM
- End: Today at 9:14 AM
- Elapsed: 8 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

nmap / Configuration
[← Back to Scan Report](#)

Settings

Credentials

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BASIC

General

Schedule

Notifications

DISCOVERY

ASSESSMENT

REPORT

ADVANCED

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nmap

Description

Folder

My Scans

Targets

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The Nessus scan was used to identify vulnerabilities and security weaknesses associated with the services discovered during Nmap enumeration.

The completed scan identified findings across multiple severity levels, including critical, high, medium, low, and informational findings.